Calcurion Physics Worksheets

A-Level Physics

Topic: Fields

MARKSCHEME

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1. $F = Gm_1m_2/r^2$ **A2** [2]
2. $F = 6.67 \times 10^{-9}$ N **M1A1** [3]
3. Force per unit mass **A1** [1]
4. $g = 9.8$ N kg^{-1} **M1A1** [3]
5. $F = kq_1q_2/r^2$ **A2** [2]
6. $F = 5.4$ N (attractive) **M1A1** [3]
7. Force per unit positive charge **A1** [1]
8. $E = 1.13 \times 10^3$ N C^{-1} **M1A1** [3]
9. $E = -\Delta V/\Delta x$ **A1** [1]
10. Work done per unit mass bringing from infinity **A1** [1]
11. $V = -6.3 \times 10^7$ J kg^{-1} **M1A1** [3]
12. $C = Q/V$ **A1** [1]
13. $C = 2.0$ μF **M1A1** [2]
14. $E = \frac{1}{2}CV^2$ **A1** [1]
15. $E = 7.2$ mJ **M1A1** [2]
16. $g = F/m = GM/r^2$ **A2** [2]
17. $v = \sqrt{GM/(R+h)}$ **A3** [3]
18. $v_{\text{esc}} = \sqrt{2GM/R} = 11.2$ km/s **M1A1** [3]
19. $E = 2.0 \times 10^4$ V m^{-1} **M1A1** [2]
20. $\frac{1}{2}mv^2 = eV$; $v = 5.9 \times 10^6$ m/s **M1A1** [4]
21. Radial lines outward **A2** [2]
22. $C = 1.2$ μF **M1A1** [2]
23. $C = 5$ μF **A1** [1]
24. $\tau = 0.10$ s; $Q = Q_0/e = 184$ μC **M1A1** [4]
25. Both inverse-square; field lines; potential **A3** [3]